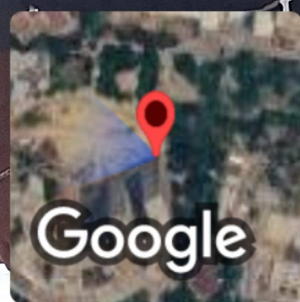
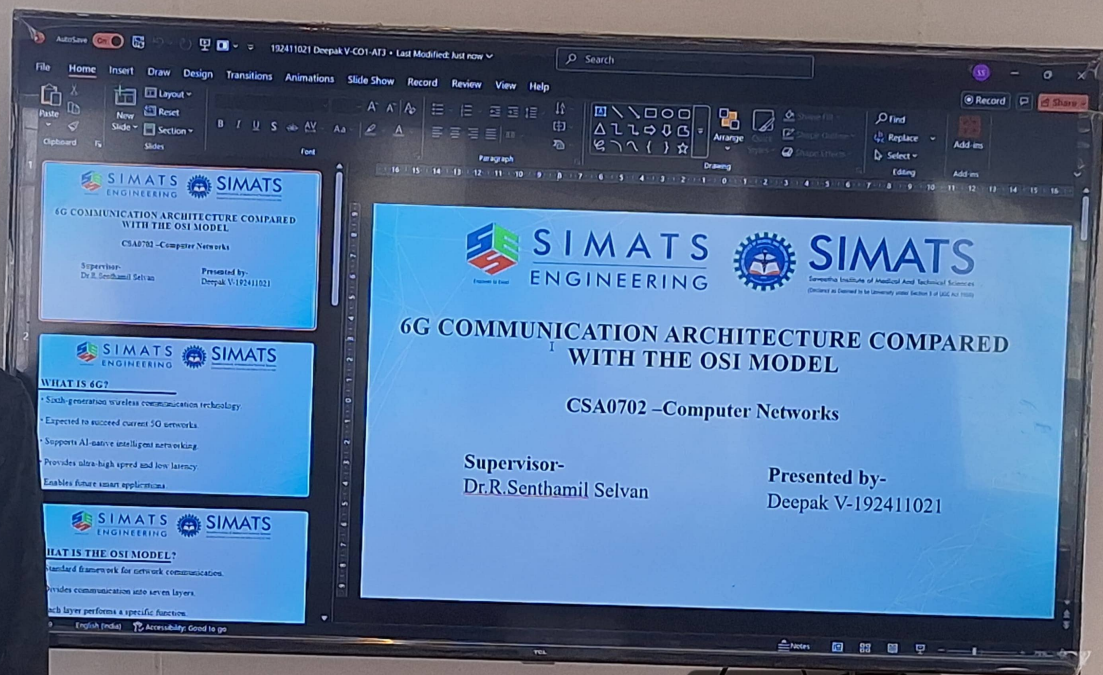
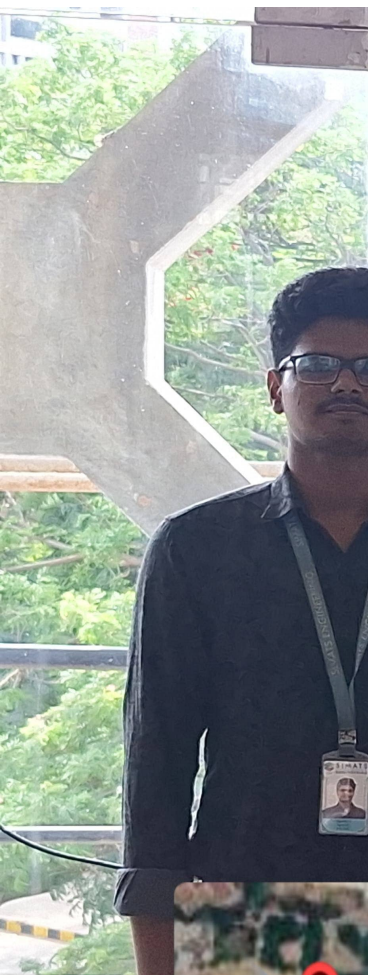




Mevalurkuppam, Tamil Nadu,
India 🇮🇳

22h8+7xq, Mevalurkuppam, Tamil Nadu 602105, India

Lat 13.02683° Long 80.017182°

Wednesday, 22/07/2026 08:55 AM GMT +05:30

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6G COMMUNICATION ARCHITECTURE COMPARED WITH THE OSI MODEL

CSA0702 –Computer Networks

Supervisor-
Dr.R.Senthamil Selvan

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WHAT IS 6G?

- Sixth-generation wireless communication technology.
- Expected to succeed current 5G networks.
- Supports AI-native intelligent networking.
- Provides ultra-high speed and low latency.
- Enables future smart applications.

WHAT IS THE OSI MODEL?

- Standard framework for network communication.
- Divides communication into seven layers.
- Each layer performs a specific function.
- Simplifies network design and troubleshooting.
- Ensures interoperability between devices.

6G COMPARED WITH THE OSI MODEL

- Physical Layer supports Terahertz communication.
- Network Layer enables AI-based routing.
- Transport Layer reduces communication latency.
- Application Layer supports immersive services.
- All layers improve intelligent communication.



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KEY FEATURES OF 6G

- Peak speed up to 1 Tbps.
- Latency below one millisecond.
- AI integrated into every network layer.
- Supports holographic communication.
- Enables massive IoT connectivity.



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APPLICATIONS

- **Smart Healthcare** – Remote robotic surgery.
- **Autonomous Vehicles** – Self-driving cars.
- **Smart Cities** – AI traffic management.
- **Extended Reality (XR)** – Apple Vision Pro.
- **Industrial Automation** – Siemens smart factories.



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ADVANTAGES

- Faster than existing 5G networks.
- Supports intelligent network management.
- Improves communication reliability.
- Reduces network response time.
- Enables next-generation applications.



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CHALLENGES

- Very high deployment cost.
- Limited Terahertz signal range.
- Security and privacy concerns.
- High energy consumption.
- Global standards are still evolving.



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CONCLUSION

- 6G represents the future of wireless communication.
- OSI Model remains the networking foundation.
- AI makes future networks more intelligent.
- 6G enables smarter and faster applications.
- Future communication depends on both technologies.